

VoltEdge Electronics — Deep Dive

Analytics review · 24-month extract 2024-07 → 2026-06 · all figures USD · prepared 2026-09

One section per business problem. Each states the decision it informs, the method on **both tracks** (DuckDB SQL + Python/pandas, parity checked), the evidence, the limitations, and a recommendation. Source files: data_analysis/parity/NN_*.py, data_analysis/findings/2026-09_NN_*.md, figures in data_analysis/outputs/.

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1 · Anchor — reproducing Net Revenue CALIBRATION

Decision. None directly — calibration before exploring, and the SQL+Python dual-track habit on a known number.

METHOD

Per docs/06: Net Revenue (USD) = SUM(net_amount_usd) – SUM(refund_amount_usd) over fact_order_lines (grain: order line), order_status <> 'cancelled'. 276,962 lines → 275,020 after removing cancelled → 167,214 distinct orders.

RESULT — BOTH TRACKS AGREE TO SUB-CENT

Metric	Value
Gross Sales (USD)	\$34,803,649
Returns (USD)	–\$3,456,572
Net Revenue (USD), full window	\$31,347,077
Current Year / Prior Year	\$22.05M / \$9.30M
By market (US / UK / BR / DE)	\$17.62M / \$4.86M / \$4.59M / \$4.28M

RECONCILIATION & CAVEATS

- The DAX formula is reproduced exactly on both tracks; Net Revenue is **\$31.35M** (\$31,347,077), the figure of record.
- **Header vs line grain:** fact_orders has 167,396 non-cancelled orders; count orders there, not on fact_order_lines (167,214). DAX-consistent AOV (Net Revenue ÷ Orders) = \$187, not the \$208 in docs/01 §12 (which is pre-returns).
- order_status in fact_order_lines only takes delivered / cancelled; returns are carried as columns on the delivered line.

2 · Growth is mostly market expansion, not the core business

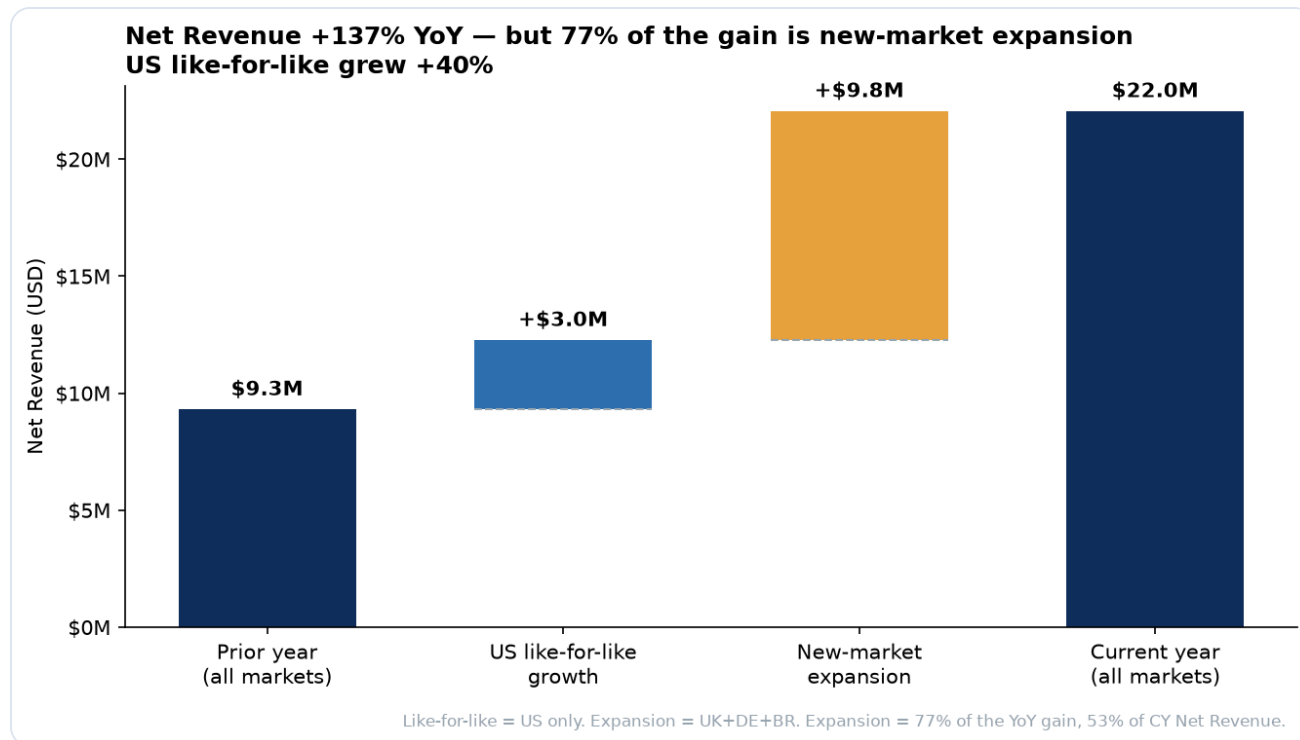
CEO / CFO

Decision. Whether to plan next year on the +137% top-line or the +40% like-for-like rate.

METHOD (BOTH TRACKS — PARITY/02_GROWTH_DECOMPOSITION.PY)

Net Revenue split CY/PY by order_date_key. **Comparable Base** =

dim_market.is_comparable_base = US only (live the whole CY+PY span), matching DAX Net Revenue (Comparable Base). **Expansion** = Total – Comparable. Plan from fact_target (metric='Net Revenue'). SQL: DuckDB FILTER (WHERE ...); Python: pandas pivot. Parity <1e-6 on every market×period cell, including the 12-month US actual-vs-plan panel used for the significance check below. Significance: one-sample t-test + bootstrap CI on monthly US attainment.



PY total → +US like-for-like → +expansion → CY total.

Cut	PY	CY	YoY
Total (all markets)	\$9.30M	\$22.05M	+137.1%
Comparable Base (US like-for-like)	\$7.33M	\$10.29M	+40.4%
Expansion (UK + DE + BR)	\$1.97M	\$11.76M	—

Expansion is 53% of CY Net Revenue and **77% of the \$12.7M YoY gain**. Actual CY vs plan: US **-2.4%**, UK +12.9%, DE +7.3%, BR -0.4%, total +1.8%.

Is the US miss real? Broken into its 12 months, US-vs-plan swings from +45% to -41% (5 months under, 7 over). Mean monthly attainment is +0.9% (positive), one-sample $t = 0.13$, $p = 0.90$, 95% CI [-12.6%, +14.2%]. The -2.4% annual figure is exact, but it is not distinguishable from ordinary month-to-month noise — it does *not* hold up as a standalone signal.

LIMITATIONS

- Comparable Base = US only (static DAX definition). A Core-3 (US+UK+DE from 2025-01) cut would give a like-for-like read from 2025 onward.
- PY UK/DE is a partial (6-month) base, so their YoY% is not meaningful — only absolute dollars are used in the decomposition.
- US +40% not yet decomposed into price/volume/mix — see §4.

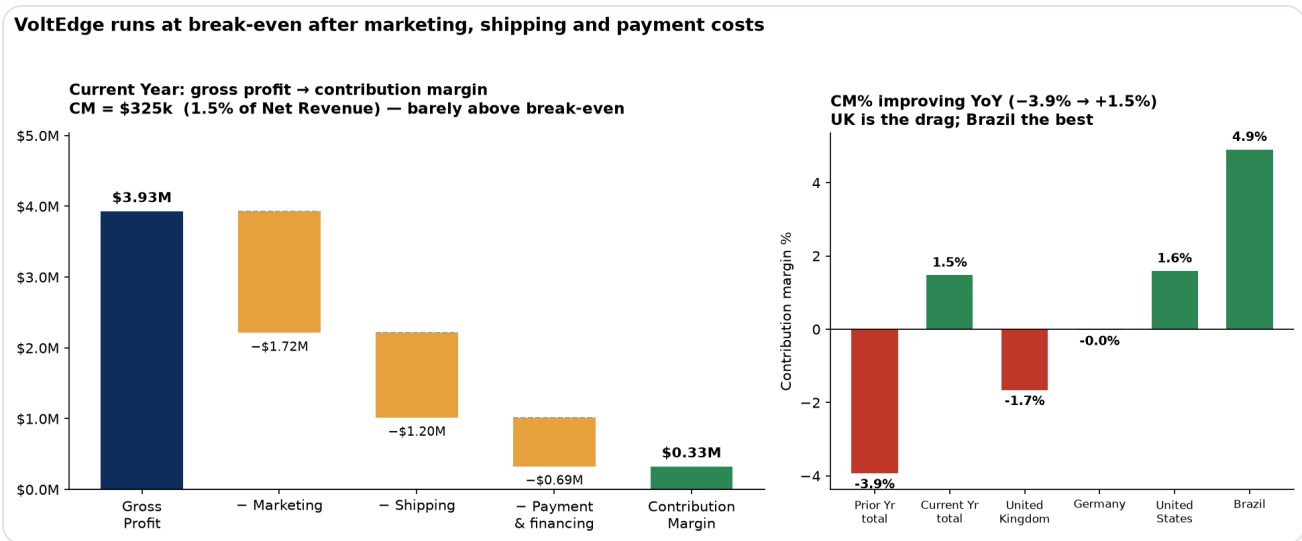
Recommendation. Report growth two ways on every board slide; anchor FY27 targets on the +40% trajectory with deceleration. Do **not** present the US -2.4% vs plan as a standalone red flag — it doesn't survive the noise check. The volatility itself ($\pm 40\%$ + monthly) is the more useful flag: build monthly plan bands wide enough to reflect it, or investigate why attainment swings that much month to month.

3 · Contribution margin: at break-even, UK the drag CFO / CEO

Decision. Whether VoltEdge can "afford" its growth, and which market destroys contribution.

METHOD (BOTH TRACKS — PARITY/03_MARGIN_TO_BREAKEVEN.PY, 30 CHECKS)

The DAX chain in docs/06: Gross Profit = Net Revenue – COGS + Returns COGS Recovered; Contribution Margin = Gross Profit – Marketing – Shipping Cost – (Payment Fees + Interest-Free Financing). Periods by order_date_key (returns by return_date_key, marketing by date_key). SQL: DuckDB CTE per component; Python: pandas.



CY gross profit → contribution margin, and CM% trajectory / by market.

	Full 24m	Current Year	Prior Year
Gross Profit	\$5.67M (18.1%)	\$3.93M (17.8%)	\$1.53M (16.5%)
– Marketing / Shipping / Payments	\$2.91M / \$1.74M / \$0.94M	\$1.72M / \$1.20M / \$0.69M	\$1.10M / \$0.54M / \$0.26M
Contribution Margin	+\$76k (0.2%)	+\$325k (1.5%)	–\$365k (–3.9%)

By market (CY): Brazil **+4.9%** (best — high gross margin + cheap Pix), US +1.6%, Germany –0.0%, UK **–1.7%** (lowest gross margin 15.4% + marketing load; *not* shipping — see §10).

LIMITATIONS

- Gross Profit adds back Returns COGS Recovered per the live DAX (pbi_measures.py); docs/06 prose line 21 says it is not — the prose is stale (+\$1.83M / +5.9pp of gross margin over the window).
- Contribution margin excludes fixed cost, payroll, tech — an operating-contribution proxy, not net profit. Real profit is materially negative.
- CM by market takes marketing at market_key face value; no HQ reallocation.

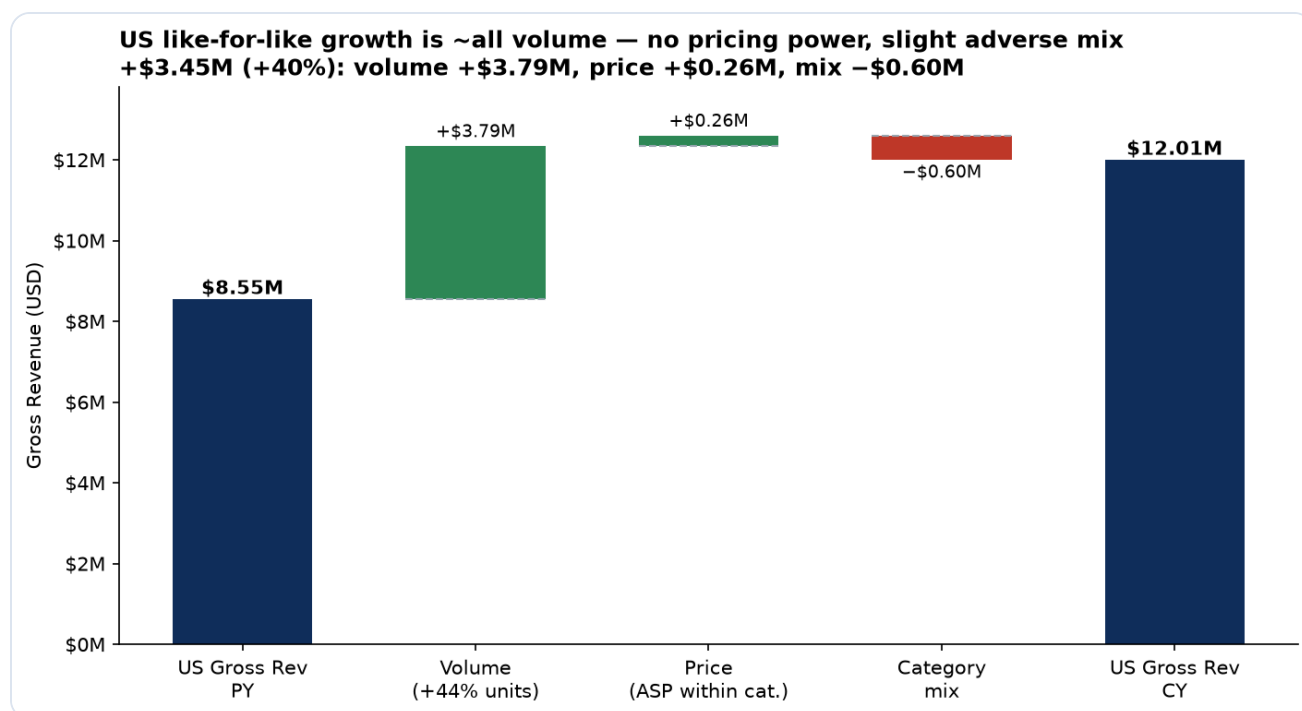
Recommendation. Board line: "positive at the contribution line for the first time (+1.5% vs -3.9%)" — but state it is \$325k, i.e. fragile. Act on the UK: lift gross margin via category mix. Protect the marketing-efficiency gain — it, not gross margin, moved CM positive (\$7).

4 · US like-for-like growth is all volume CEO / CFO / VP E-COM

Decision. Whether the US engine can carry the business to profit — a volume-only, flat-margin expansion does not.

METHOD (BOTH TRACKS — PARITY/04_PRICE_VOLUME_MIX_US.PY, 7 CHECKS)

Price/Volume/Mix per docs/06 §07, US non-cancelled fact_order_lines, by dim_product.category. Price = $\sum (\text{ASP}_{\text{CY}} - \text{ASP}_{\text{PY}}) \times \text{Units}_{\text{PY}}$; Volume = $\text{GrossRev}_{\text{PY}} \times \text{unit growth}$; Mix = $\Delta \text{GrossRev} - \text{Price} - \text{Volume}$. Parity exact.



US Gross Revenue PY → CY: volume +\$3.79M, price +\$0.26M, mix -\$0.60M.

US Gross Revenue \$8.55M → \$12.01M (+40.3%); units +44.3%. **Volume +\$3.79M (110% of the gain)**, price +\$0.26M (+8%), **mix -\$0.60M (-17%)**. Revenue share is shifting **toward Smartphones (+2.7pp)** — thinnest gross margin (8–14%) — and away from Laptops (-2.0pp).

LIMITATIONS

- ASP numerator (Gross Revenue) includes warranty lines while the denominator (Units) is product-only — the DAX definition; consistent across CY/PY so the effect split holds.
- PVM is on Gross Revenue, not margin dollars; a margin-weighted mix effect would be more negative given the Smartphone shift.
- Single comparable market (US).

Recommendation. A deliberate mix-up plan into audio / accessories / wearables is worth ~1–2pp of margin on a \$12M base. Set a category-mix guardrail in the exec pack; do not

assume operating leverage from volume alone.

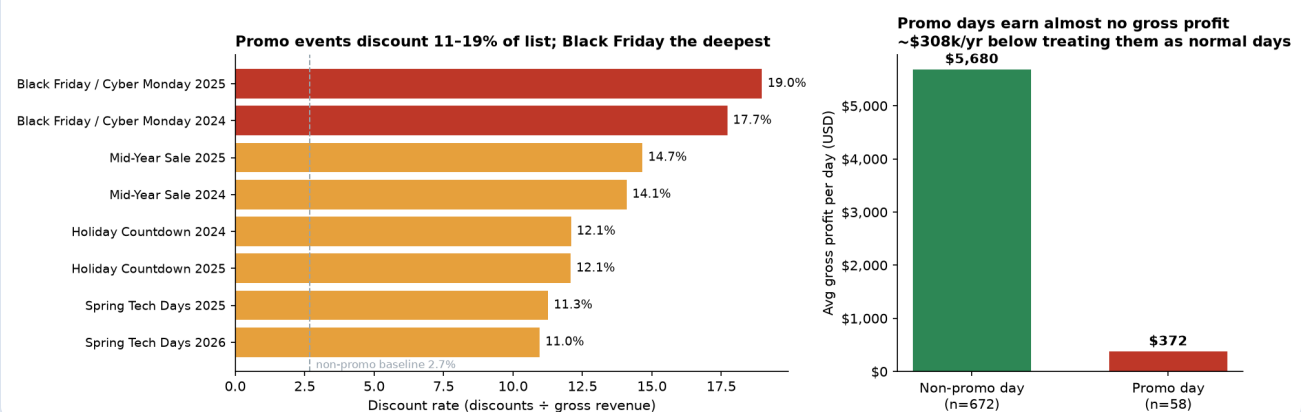
5 · Promo depth erases gross profit CFO / CMO

Decision. Keep the promo calendar as-is, or cut discount depth / shorten events.

METHOD (BOTH TRACKS — PARITY/05_DISCOUNT_LEAKAGE.PY)

Discount Rate % = $\text{SUM}(\text{discount_amount_usd}) \div \text{SUM}(\text{gross_amount_usd})$. Promo days from `dim_date.is_promo_period`. Promo-day P&L uses line-level $\text{gp} = (\text{net} - \text{refund}) - \text{cogs_usd}$ aggregated to a per-day average. Discount-rate parity exact to $1\text{e-}9$; per-day gross profit parity (sum/mean/variance per group) before any significance test runs on it. Significance: Welch's t-test. Uncertainty: percentile bootstrap (10,000 resamples, seed 42). Seasonality check: OLS with calendar-month fixed effects, robust SE.

Discounting overall is modest (4.6%) — the problem is promo depth, not creep



Discount rate by promo event; average daily gross profit promo vs non-promo.

- Blended discount rate **4.6%** — low end of the 4–10% benchmark; mild drift PY 4.3% → CY 4.7%. Non-promo days 2.7%; **promo days 15.7%**.
- Promo day: $\sim 2\times$ the volume of a normal day but **\$372 of gross profit vs \$5,680** (Welch $t = -8.80$, $p = 1.2 \times 10^{-12}$). Annual impact $\approx$ **-\$308k** [95% CI: $-\$379\text{k}$, $-\$242\text{k}$] across 58 promo days — roughly the size of the entire CY contribution margin.
- Black Friday is the deepest (17–19% off); discount rate is flat across categories, so this is a calendar/depth decision, not a merchandising one.

LIMITATIONS

- Pull-forward not modelled (would make promos look *worse*).
- New-customer acquisition & inventory-clearance value not credited (could justify part of the cost — needs \$8 and an aged-inventory view).
- Baseline-day GP is a blended average, and promo days concentrate in specific calendar months. Tested directly (month-fixed-effects regression): the seasonality-adjusted gap is $-\$5,710/\text{day}$, essentially unchanged from the naive $-\$5,308$ — **not** a seasonality artifact.

Recommendation. Cap Black Friday at 12–14% and test shorter windows; re-scope other events toward margin-carrying categories. If BF is kept, fund it from the marketing budget with a CAC target, not invisibly from gross margin. Add "gross profit per promo day vs baseline" to the CFO pack.

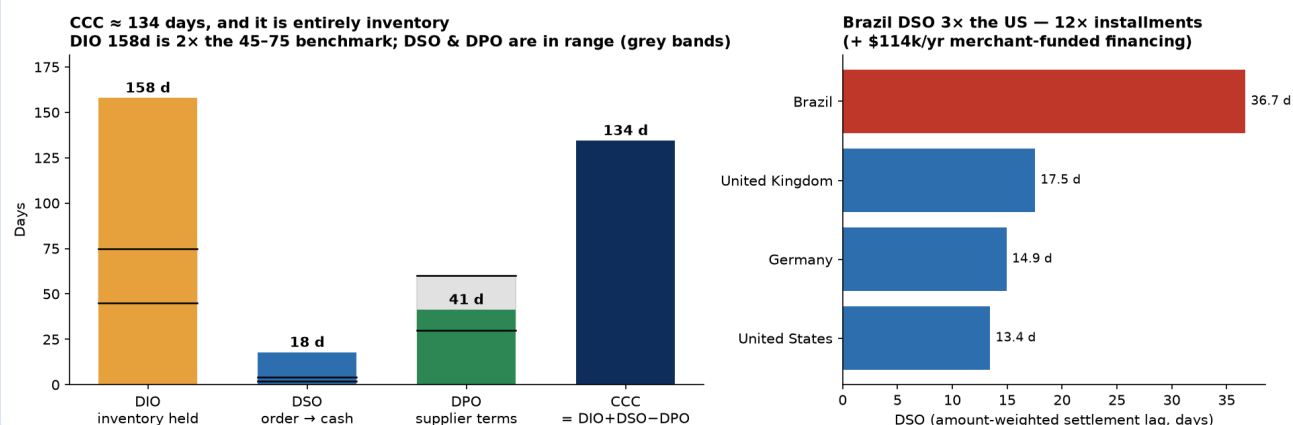
6 · Working capital: ~\$4.4M frozen in inventory CFO / COO

Decision. Which lever — inventory, receivables, or payables — releases cash. The answer is inventory.

METHOD (BOTH TRACKS — PARITY/06_CASH_CONVERSION_CYCLE.PY)

Per docs/06 §11: DSO = amount-weighted settlement lag over fact_payment_schedule; DIO = Avg Inventory Value $\times$ 365 $\div$ COGS_{12M} (last-12 snapshot months); DPO = AP Open $\times$ 365 $\div$ COGS_{12M} using the mean month-end AP over the last 12 (the DAX point-in-time value is a noisy trough at 2026-06-30). CCC = DIO + DSO - DPO. Parity exact on every component.

The working-capital problem is bigger than the P&L: ~\$4.4M cash could be freed from inventory



CCC composition vs benchmark bands; DSO by market.

Metric	VoltEdge	Benchmark
DIO	158 days	45–75
DSO	17.8 days	US/EU low, BR higher
DPO	41 days	30–60
CCC	$\approx$ 134 days	lower is better

Average inventory rose from \$5.85M (24m) to **\$8.31M (last 12m)**. At DIO 75 it would be $\sim$ \$3.95M $\rightarrow$ **$\approx$ \$4.4M of cash freed**. Payables run at terms (paid $\sim$ 1 day past due) — no easy cash there. Brazil DSO 36.7 days (12 $\times$ installments) + \$114k/yr merchant-funded "sem juros" financing; on current numbers a good trade for +\$4.6M revenue and the best CM.

LIMITATIONS

- DIO basis matters: last-12m gives 158, full-24m gives 111 — both well above benchmark; pick one.
- Month-end AP is a modelled proxy for a true daily AP balance.

- No cost-of-capital rate applied — the \$4.4M is gross cash, not an interest saving.

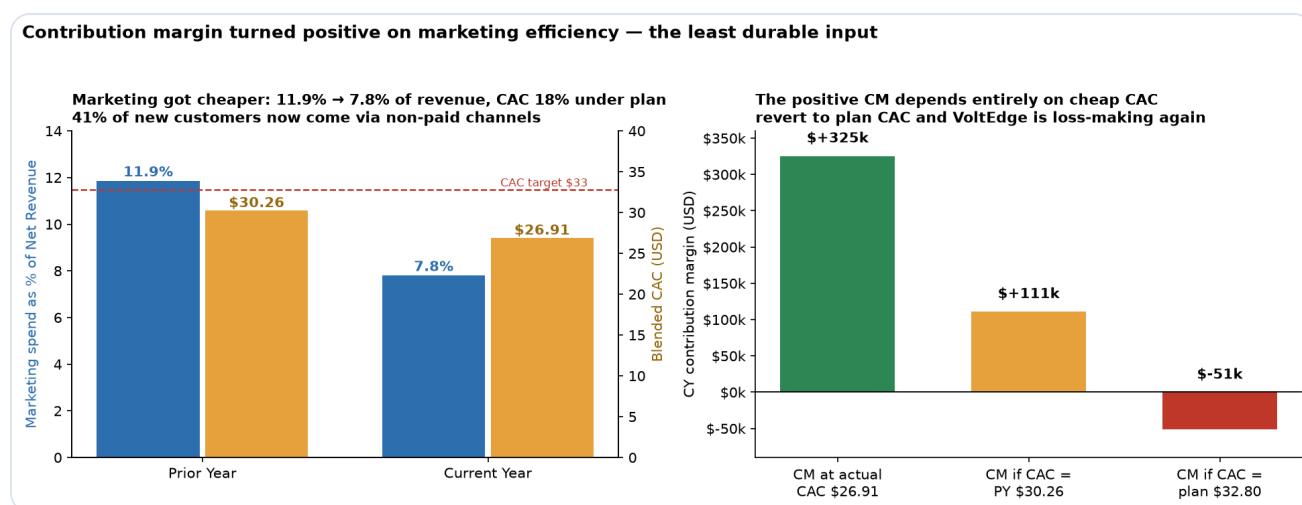
Recommendation. CFO priority #1: an inventory-reduction program — SKU-level reorder points, kill dead stock (\$9) — targeting DIO 90 within two quarters. Frees ~\$3M immediately. Do not extend DPO further. Put CCC and Inventory Value trend on the Executive Summary.

7 · Contribution margin depends on cheap CAC CFO / CMO

Decision. How much confidence to place in the "we reached break-even" message.

METHOD (BOTH TRACKS — PARITY/07_MARKETING_EFFICIENCY.PY)

docs/06 §08: New Customers = customers whose first-ever non-cancelled order (MIN(order_date_key)) is in the period; Blended CAC = Spend ÷ New Customers; MER = Net Revenue ÷ Spend. Sliced full/CY/PY, by channel group, vs fact_target. SQL: DuckDB view on first order; Python: pandas. Parity exact, including a 24-row monthly spend/new-customer panel. Significance: paired t-test (calendar month PY vs CY) + bootstrap CI; log-log OLS for the spend→new-customers association.



Marketing intensity & CAC PY vs CY; CY contribution-margin sensitivity to CAC.

	PY	CY
Blended CAC	\$30.26	\$26.91 (target \$32.80 → -18%)
MER (Net Rev ÷ spend)	8.4×	12.8× (benchmark 6–10×)
Marketing as % of Net Revenue	11.9%	7.8%

The -4.1pp drop in marketing intensity ≈ the entire +5.4pp CM swing. Channel mix did *not* shift — 96.7% of spend is paid both years; what changed is efficiency (41% of CY new customers now arrive via non-paid channels; paid-only CAC is \$44.26). **CY contribution margin: +\$325k at actual CAC; +\$111k at PY CAC; -\$51k at plan CAC.**

Is the CAC improvement itself real? Paired by calendar month (12 pairs), the mean CAC change is -\$1.18, $p = 0.56$, 95% CI [-\$4.64, +\$2.61] — spans zero. The annual figures are exact; whether the gap is a durable efficiency gain or a few unusually cheap months is not established, which makes the case for treating CM as fragile *stronger*, not weaker. Separately, a spend↔new-customer

elasticity of **1.01 [0.53, 1.49]** (observational) shows no sign yet of diminishing returns on paid spend.

LIMITATIONS

- Left-censoring: dataset starts 2024-07, so PY/full new-customer revenue reads as 100%; only the CY split (84% new / 16% returning by the DAX definition) is meaningful — reconciled with the 44% order-month figure in §8.
- ROAS/attribution is last-touch at channel level, not an MMM.
- CM-sensitivity flexes only CAC, holding new-customer count fixed — a cushion estimate, not a forecast.
- The elasticity is observational (spend and organic demand share a seasonal calendar), not a causal "spend +1%" estimate.

Recommendation. Reframe the board message to "positive at current CAC — and the improvement funding it isn't yet shown to be durable". Track marginal CAC monthly against the paired-month baseline; stress-test at \$33 and \$45. Given MER 12.8× and an elasticity near 1, test scaling paid spend to a marginal-CAC ceiling (~\$45) — there is likely profitable volume above budget.

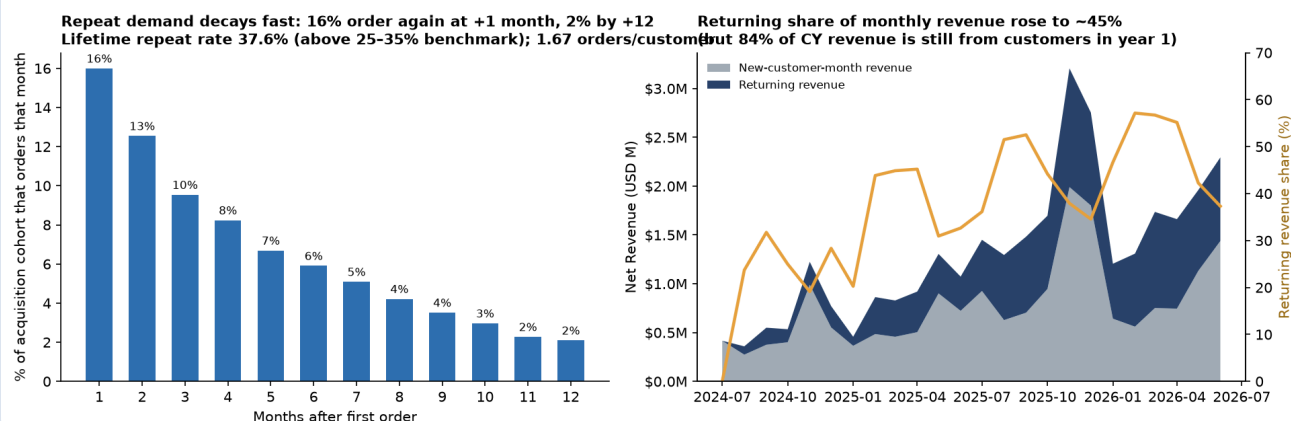
8 · Retention: healthy, not yet the engine VP CUSTOMER / CEO

Decision. Whether "repeat revenue gets us to profit" is credible, and where retention spend should go.

METHOD (BOTH TRACKS — PARITY/08_RETENTION_COHORTS.PY)

Repeat Purchase Rate % = (customers with >1 non-cancelled order) ÷ (customers with ≥1). Cohort retention: acquisition month = month of first order; share of the cohort ordering again at month-offset n , averaged over cohorts with ≥6 months maturity. New-vs-returning revenue at (customer, calendar-month) grain. Repeat rates parity-exact to 1e-9.

Repeat behaviour is healthy and improving — but not yet the revenue engine



Cohort re-order curve and new-vs-returning revenue by month.

- Lifetime repeat rate **37.6%** (above the 25–35% benchmark); 1.67 orders/customer. Newer markets out-repeat the US: DE 40.5%, BR 40.2%, UK 40.0%, **US 36.0%**.
- Re-order decays fast: 16% at +1 month, 9.5% at +3, 2.1% at +12. Only 26.5% of buyers are "Active".
- Two "returning revenue" numbers, both correct: DAX New Revenue % (customer "new" for the whole first year) → 16% returning; order-month grain → 44% returning. The gap is the within-year repeat orders of customers acquired this year.

LIMITATIONS

- Left-censoring flatters newer-market repeat rates and depresses the tail of the average curve.
- Retention curve is order-count based; a revenue curve decays slightly slower.

Recommendation. Own a +1-to-+3-month reactivation program (lifecycle CRM, accessory cross-sell into margin-carrying categories); target +1-month re-order 16% → 20%. Report both returning-revenue definitions with the reconciliation. Investigate why US repeat lags the newer markets by 4pp.

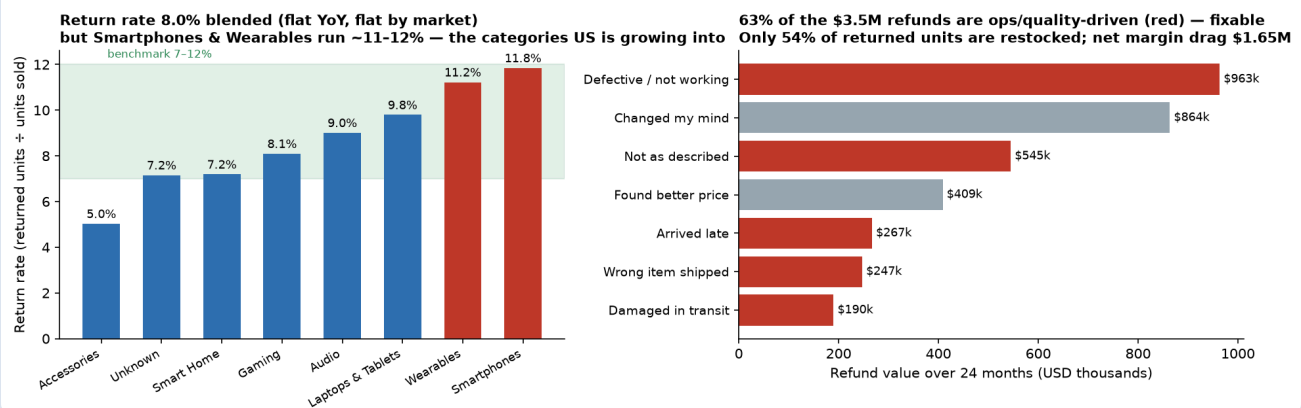
9 · Returns: an \$1.65M margin leak, mostly fixable VP CUSTOMER / COO

Decision. Whether returns need a dedicated program, and where.

METHOD (BOTH TRACKS — PARITY/09_RETURNS.PY)

Return Rate % = $\text{SUM}(\text{returned_qty}) \div \text{Units Sold (product lines)}$. Net margin drag = refund – cogs_recovered_usd. Sliced by category, market, reason. Rate & value parity exact.

Returns are not rising — but \$1.65M of margin leaks, mostly for fixable reasons



Return rate by category vs benchmark; refund value by reason (controllable vs choice).

- Return rate 8.0% — mid-benchmark, flat YoY and across all four markets.
- **Net margin drag \$1.65M** over 24m (refund \$3.48M – COGS recovered \$1.83M) = 5.3% of Net Revenue, 29% of gross profit.
- Restock rate only 54% — 46% of returned units are written down. Biggest lever on the drag.
- Smartphones 11.8% and Wearables 11.2% (highest) — the categories US growth concentrates into. **63% of refunds are ops/quality-driven** (defective \$963k, "not as described" \$545k, wrong item, damaged, late).

LIMITATIONS

- fact_returns refund total (\$3.48M) is ~1% above the fact_order_lines refund column used for Net Revenue — a few returns map to cancelled/re-adjusted orders.
- "Controllable vs choice" is an analyst bucketing of the 7 reason codes.
- Unit-based rate; a value-based rate would be higher for Smartphones/Laptops.

Recommendation. Two KPIs on the Ops/CRM page: restock rate (54% → 70%, ≈ +\$180k per +10pp) and controllable return rate. Prioritise QA + PDP content on Smartphones and Wearables. Combined realistic recovery ≈ \$0.4–0.7M/yr.

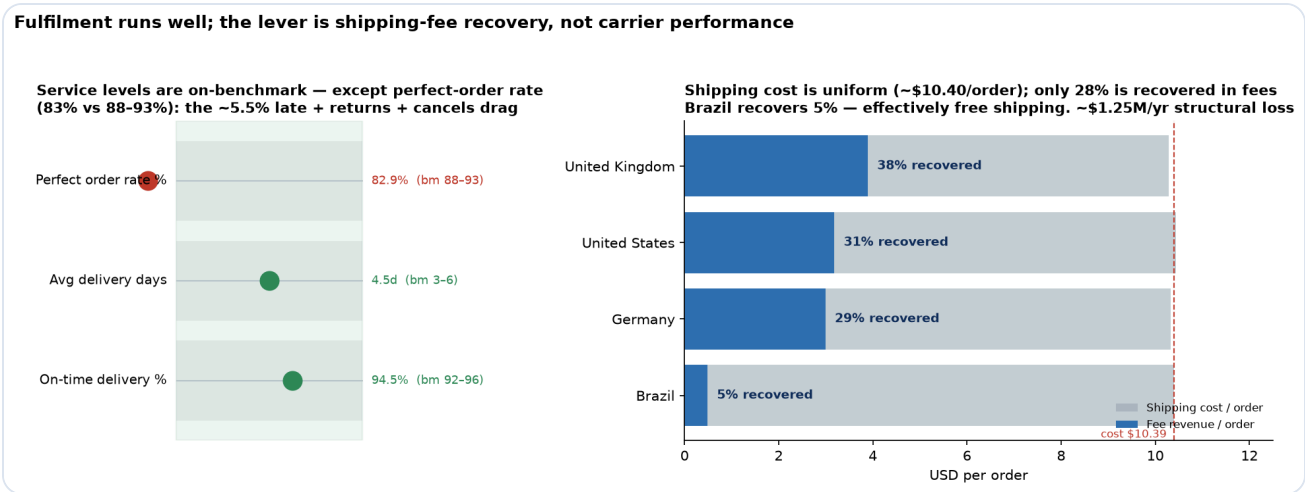
10 · Fulfilment & shipping-fee recovery COO / CFO

Decision. Where (if anywhere) to invest in logistics.

METHOD (BOTH TRACKS — PARITY/10_FULFILLMENT.PY)

docs/06 §10 KPIs on non-cancelled fact_orders, sliced by market, carrier, Q4 vs rest. SQL:

DuckDB COUNT(DISTINCT ...) FILTER; Python: pandas. Parity exact on every KPI and market cell.



Service levels vs benchmark; shipping cost vs fee recovery by market.

KPI	VoltEdge	Benchmark
On-time delivery %	94.5%	92–96%
Avg delivery days	4.5	3–6
Perfect order rate %	82.9%	88–93%
Shipping cost recovery %	28%	—

Service levels are uniform and on-target; carrier spread is tiny. Shipping costs \$1.74M/24m; fees recover ~\$0.49M → ~\$0.6M/yr structural loss. Recovery: UK 38%, US 31%, DE 29%, BR 5%. Q4 cost/order +12% — manageable. **Corrects §3:** the UK CM drag is gross margin, not shipping (UK ship cost/order is the lowest of the four).

LIMITATIONS

- Synthetic carrier data shows little differentiation — don't over-read "no bad carrier".
- Free-shipping-threshold logic is embedded in the fee, not modelled separately.

Recommendation. CFO/CMO joint call on the free-shipping threshold market by market, starting with Brazil. Lifting blended recovery 28% → 45% ≈ +\$0.3–0.4M/yr contribution. Do

not spend on carrier changes. Track shipping cost recovery % on the Executive Summary alongside CCC.

Cross-finding synthesis

The findings compound. VoltEdge's reported +137% growth is mostly one-time expansion (§2); the underlying US engine grows +40% but purely on volume into its thinnest-margin, highest-return categories (§4, §9), so gross margin barely moves. Contribution margin turned positive (§3) almost entirely because CAC came in 18% under plan (§7) — revert to plan and it is negative again. Two controllable leaks (promo depth \$5 ≈ \$0.3M/yr, shipping subsidy \$10 ≈ \$0.6M/yr) are each comparable to the whole contribution margin. And the largest prize sits on the balance sheet: ~\$4.4M of cash frozen in an oversized inventory (§6). The credible long-run profit path is repeat revenue (§8) — healthy at 37.6% but too young to carry the plan. **Priority order: (1) working-capital / inventory program, (2) protect & stress-test CAC, (3) category mix-up + returns cost, (4) promo & shipping-fee discipline, (5) fund +1–3 month retention.**

All numbers reproduced on two independent stacks (DuckDB SQL + Python/pandas) from data/curated/; parity scripts in data_analysis/parity/, per-finding write-ups in data_analysis/findings/.